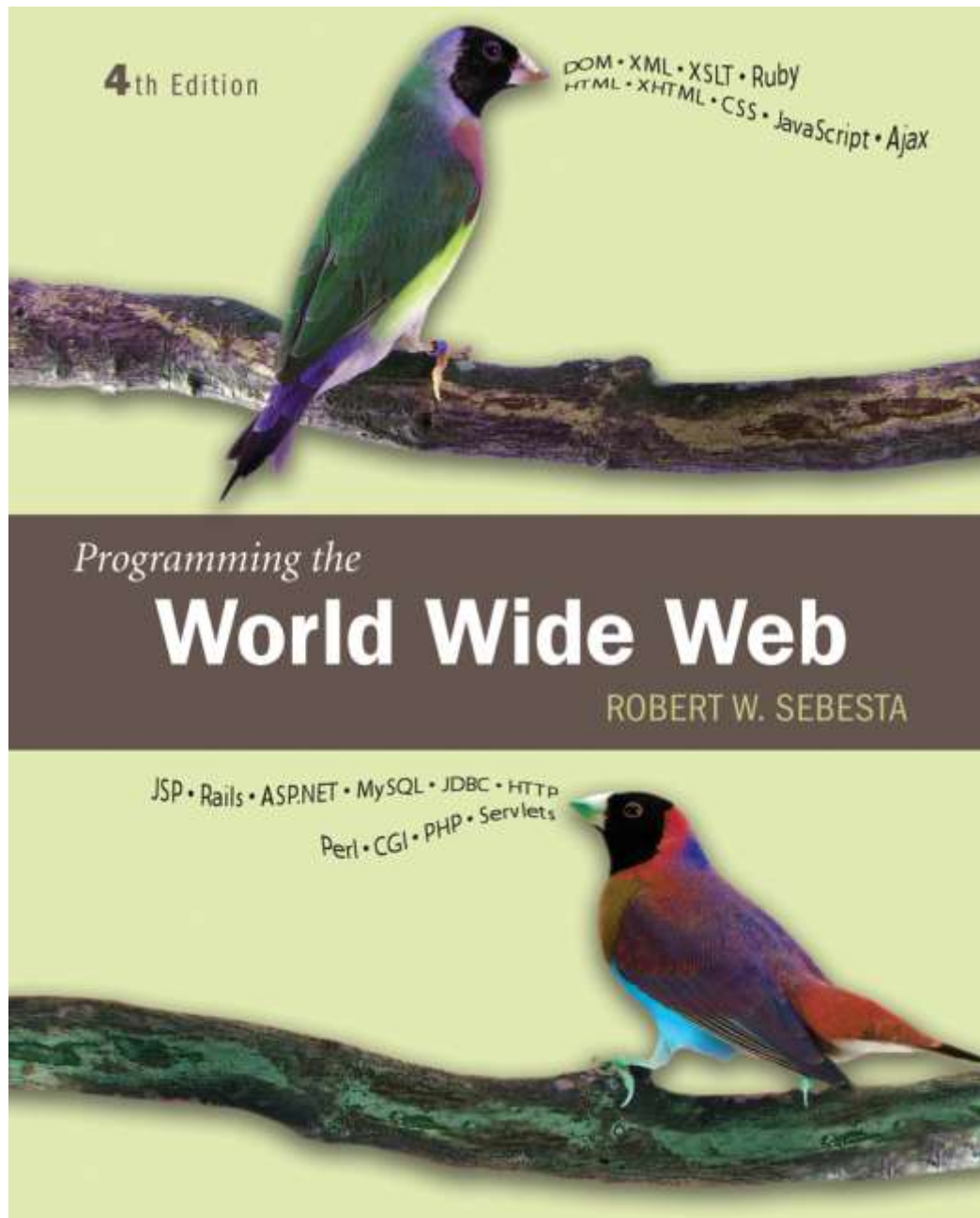




Chapter 1

Fundamentals

1.1 A Brief Intro to the Internet

- **Internet History**
- **Internet Protocols**

1.1 Internet History

- Origins
 - ARPAnet - late 1960s and early 1970s
 - Network reliability
 - For ARPA-funded research organizations
- BITnet, CSnet - late 1970s & early 1980s
 - email and file transfer for other institutions
- NSFnet - 1986
 - Originally for non-DOD funded places
 - Initially connected five supercomputer centers
 - By 1990, it had replaced ARPAnet for non-military uses
 - Soon became the network for all (by the early 1990s)
- NSFnet eventually became known as the Internet

1.1 Internet History

- What the Internet is:
 - A world-wide network of computer networks
 - At the lowest level, since 1982, all connections use TCP/IP
 - TCP/IP hides the differences among devices connected to the Internet

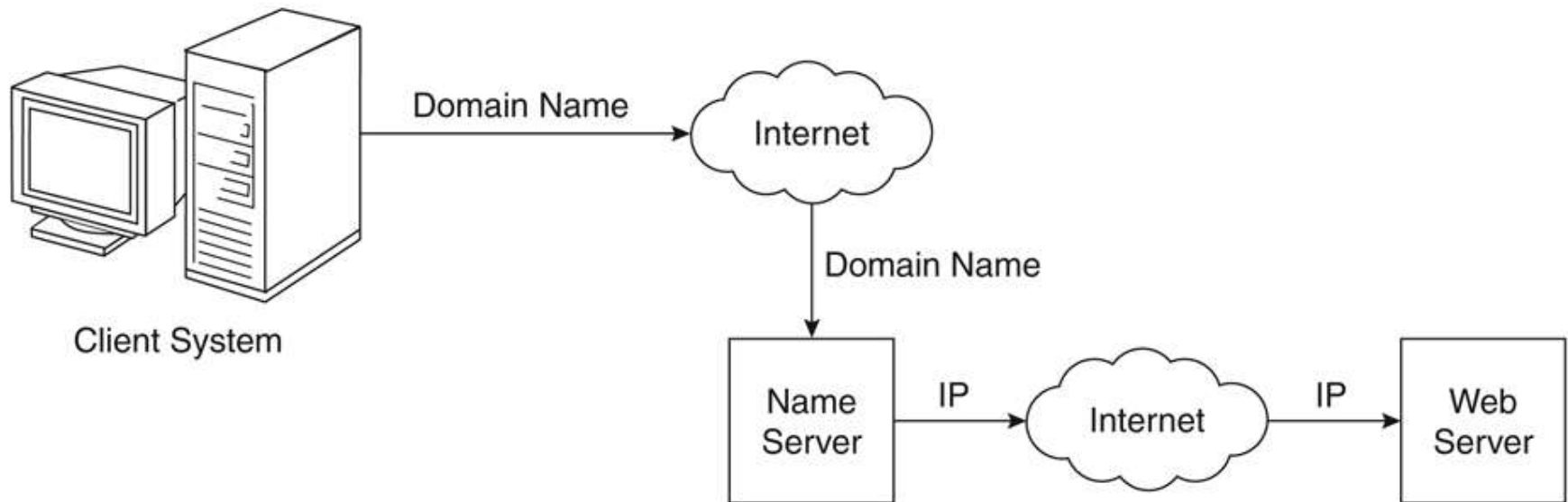
1.1 Internet Protocols

- Internet Protocol (IP) Addresses
 - Every node has a unique numeric address
 - Form: 32-bit binary number
 - New standard, IPv6, has 128 bits (1998)
- Organizations are assigned groups of IPs for their computers
- Problem: By the mid-1980s, several different protocols had been invented and were being used on the Internet, all with different user interfaces (Telnet, FTP, Usenet, mailto)

1.1 Internet Protocols

- Domain names
 - Form: host-name.domain-names
 - First domain is the smallest; last is the largest
 - Last domain specifies the type of organization
 - Fully qualified domain name - the host name and all of the domain names
 - DNS servers - convert fully qualified domain names to IPs

Figure 1.1 Domain name conversion



Client and Server

- **Clients and Servers are programs that communicate with each other over the Internet**
- **A Server runs continuously, waiting to be contacted by a Client**
 - **Each Server provides certain services**
 - **Services include providing web pages**
- **A Client will send a message to a Server requesting the service provided by that server**
 - **The client will usually provide some information, parameters, with the request**

1.2 The World-Wide Web

- **A possible solution to the large number of different protocols being used on the Internet**
- **Origins**
 - **Tim Berners-Lee at CERN proposed the Web in 1989**
 - **Purpose: to allow scientists to have access to many databases of scientific work through their own computers**
 - **Document form: hypertext**
 - **Pages? Documents? Resources?**
 - **We'll call them documents**
 - **Hypermedia – more than just text – images, sound, etc.**

1.2 The World-Wide Web

- **Web or Internet?**
 - **The Web uses one of the protocols, http, that runs on the Internet--there are several others (telnet, mailto, etc.)**

1.3 Web Browsers

- **Browsers are clients - always initiate, servers react (although sometimes servers require responses)**
- **Mosaic - NCSA (Univ. of Illinois), in early 1993**
 - **First to use a GUI, led to explosion of Web use**
 - **Initially for X-Windows, under UNIX, but was ported to other platforms by late 1993**
- **Most requests are for existing documents, using HyperText Transfer Protocol (HTTP)**
 - **But some requests are for program execution, with the output being returned as a document**
- **Name some of the Web Browsers?**

1.4 Web Servers

- **Provide responses to browser requests, either existing documents or dynamically built documents**
- **Browser-server connection is now maintained through more than one request-response cycle**
- **All communications between browsers and servers use Hypertext Transfer Protocol (HTTP)**
- **Port number for HTTP? and HTTPS?**

1.4 Web Server Operation

- **Web servers run as background processes in the operating system**
 - **Monitor a communications port on the host, accepting HTTP messages when they appear**
- **All current Web servers came from either**
 - 1.The original from CERN**
 - 2.The second one, from NCSA**
- **Most common server now is Apache**

1.4 Web Server Operation Details

- The file structure of Web servers have two main directories:
 - 1.Document root** (servable documents)
 - 2.Server root** (server system software)
- Document root is accessed indirectly by clients
 - Its actual location is set by the server configuration file
 - Requests are mapped to the actual location
- **Virtual document trees** (servable documents from the secondary areas)
- **Virtual hosts** (servers support more than one site on a computer)
- **Proxy servers** (Servers serving documents from the document root of other machine)
- Web servers now support other Internet protocols

1.4 Web Server Operation : Apache

- Apache (open source, fast, reliable)
- 3 config files of Apache server:
 - Httpd.conf
 - Srm.conf
 - Access.conf
- Different directives are available in <http://httpd.apache.org/docs/2.2/mod/quickreference.html>
- Basically designed for UNIX systems

1.4 Web Server Operation : IIS

- IIS for windows systems
 - Operation is maintained through a program (IIS snap-in) with a GUI interface
 - Operations of Apache is controlled by configuration files

1.5 URLs

- General form:

Format: object-address

- The scheme is often a communications protocol, such as telnet or ftp
- For the http protocol, the object-address is:
//fully-qualified-domain-name/path-to-document
- For the file protocol, only the doc path is needed
file://path-to-document

1.5 URLs

- Host name may include a port number, as in zeppo:80 (80 is the default, so this is silly)
- URLs cannot include spaces or any of a collection of other special characters (semicolons, colons, ...)
- The doc path may be abbreviated as a partial path
 - The rest is furnished by the server configuration
- If a special character has to be entered then % with its hexadecimal ASCII code should be included

Eg: if domain name '**RV CE**' has to be specified then it has to be written as '**RV%20CE**' (**20 is ASCII code for space**)

1.6 Multipurpose Internet Mail Extensions (MIME)

- Used by the browsers for deciding the format of doc received by the web server
- Originally developed for email
- Attached by the server to the beginning of the document
- Type specifications
 - Form:
type/subtype
 - Common types are: text, image and video
 - Common subtypes are: plain, html, gif, jpeg, mpeg, quicktime
 - Examples: text/plain, text/html, image/gif, image/jpeg

1.6 Multipurpose Internet Mail Extensions (MIME)

- Server gets type from the requested file name's suffix (.html implies text/html)
- Browser gets the type explicitly from the server
- Experimental sub types
- Subtype begins with x-
 - e.g., video/x-msvideo
- Experimental types require the server to send a helper application or plug-in so the browser can deal with the file

1.7 The HyperText Transfer Protocol

- The protocol used by ALL Web communications
- Request Phase

- Form:

HTTP method domain part of URL HTTP ver.

Header fields

blank line

Message body

- An example of the first line of a request:

GET /degrees.html HTTP/1.1

1.7 The HyperText Transfer Protocol: Methods

- GET - Fetch a document
- POST - Execute the document, using the data in body
- HEAD - Fetch just the header of the document
- PUT - Store a new document on the server
- DELETE - Remove a document from the server

1.7 HTTP Headers

- Four categories of header fields:

General, request, response, & entity

- Common request fields:

Accept: text/plain

Accept: text/*

If-Modified_since: date

- Common response fields:

Content-length: 488

Content-type: text/html

- Can communicate with HTTP without a browser

> telnet blanca.uccs.edu http

GET /respond.html HTTP/1.1

Host: blanca.uccs.edu connection:keep-alive ---- ----- ----- hello world

1.7 HTTP Response

- Form:
 - Status line
 - Response header fields
 - blank line
 - Response body
- Status line format:
 - HTTP version status code explanation
- Example: HTTP/1.1 200 OK
 - (Current version is 1.1)
- Status code is a three-digit number; first digit specifies the general status
 - 1 => Informational
 - 2 => Success
 - 3 => Redirection
 - 4 => Client error
 - 5 => Server error
- The header field, Content-type, is required

1.7 HTTP Response Example

HTTP/1.1 200 OK

Date: Tues, 18 May 2004 16:45:13 GMT

Server: Apache (Red-Hat/Linux)

Last-modified: Tues, 18 May 2004 16:38:38 GMT

Accept-ranges: bytes

Content-length: 364

Connection: close

Content-type: text/html, charset=ISO-8859-1

- Both request headers and response headers must be followed by a blank line

1.7 Status code Example

- Example for status code are
 - ❖ 404 --- file not found
 - ❖ 200 --- OK(Successful Request)
 - ❖ 500 --- Server encountered a problem
 - ❖ 202- Accepted for processing

1.8 Security

- Both Web **Browsers** and **Servers** are unsafe !!!??
- Security issues are
 1. **Privacy**: it shouldn't be possible to steal the data while transmitting
 2. **Integrity**: it shouldn't be possible to modify the data while transmitting
 3. **Authentication**: it should be possible for both ends to be certain of each others identity
 4. **Nonrepudiation**: it should be possible to legally prove that the message was actually sent and received
- **Encryption** is the tool for resolving security issues
 1. Symmetric key encryption
 2. Public key encryption

1.8 Security

- **Some more security problems are**
 - 1. DoS**
 - 2. Viruses**
 - 3. Worms**